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Chapter 42 **42.1** John Cancalosi/Alamy Stock Photo; **42.2a** Reinhard Dirscherl/WaterFrame/Getty Images; **42.2b** Eric Grave/Science Source; **42.9** **top** Indigo Instruments; **bottom** Ed Reschke/Photolibrary/Getty Images; **p. 931** **Yanagisawa** Skeeter Hagler; **42.18** Eye of Science/Science Source; **42.19** Image Source Plus/Alamy Stock Photo; **Scientific Skills Exercise** cassis/Fotolia; **42.21a** Peter Batson/Image Quest Marine; **42.21b** Olgysya/Shutterstock; **42.21c** Greg Amptman/Shutterstock; **42.23c** Prepared by Dr. Hong Y. Yan, University of Kentucky and Dr. Peng Chai, University of Texas; **42.24** Motta and Macchiarrelli, Anatomy Dept., Univ. La Sapienza, Rome/Science Source; **42.26** Hans-Rainer Duncker, Institute of Anatomy and Cell Biology, Justus-Liebig-University Giessen; **42.32** Doug Allan/Nature Picture Library; **p. 951** **spider** CB2/ZOB/WENN.com/Newscom.

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Chapter 47 **47.1** Brad Smith/Stamps School of Art & Design, University of Michigan; **inset** Oxford Scientific/Getty Images; **47.3** **top** Victor D. Vacquier; **bottom** From: Wave of free calcium at fertilization in the sea urchin egg visualized with fura-2. M. Hafner et al. *Cell Motil Cytoskeleton*. 1988;9(3):271-7. Fig. 1; **47.6** George von Dassow; **47.7** **top** Jürgen Berger/Max Planck Institute for Developmental Biology, Tübingen Germany; **bottom** Andrew J. Ewald, Johns Hopkins Medical School; **47.13b** Alejandro Díaz Diez/AGE Fotostock/Alamy Stock Photo; **47.14a** P. Huw Williams and Jim Smith, The Wellcome Trust/Cancer Research UK Gurdon Institute; **47.14c** Thomas Poole, SUNY Health Science Center; **47.15b** Keith Wheeler/Science Source; **47.18b** From: Cell lineage analysis in ascidian embryos by intracellular injection of a tracer enzyme. III. Up to the tissue restricted stage. H. Nishida. *Dev Biol*. 1987 Jun;121(2):526-41. Fig. 1. Reprinted by permission of Academic Press; **47.19** From: Post-embryonic cell lineages of the nematode, *Caenorhabditis elegans*. E. Sulston et al. *Dev Biol*. 1977 Mar;56(1):110-56. Fig. 1; **47.20** Susan Strome; **47.21** Susan Strome; **47.24** From: Dorsal-ventral patterning and neural induction in *Xenopus* embryos. E. M. De Robertis and H. Kuroda. *Annu Rev Cell Dev Biol*. 2004;20:285-308. Fig. 1; **47.25a** Kathryn Tosney, University of Michigan; **47.26** Based on Honig and Summerbell, courtesy of Lawrence S. Honig; **p. 1066** **turtle** James Gerholdt/Getty Images.

Chapter 48 **48.1** Franco Banfi/Science Source; **48.2** Edwin R. Lewis; **p. 1068** **Olivera** From: QnAs with Baldomero M. Olivera. Beth Azar. *PNAS* August 21, 2012 109 (34) 13470; <https://doi.org/10.1073/pnas.1211581109>. Fig. 1; **48.5** Thomas Deerinck/National Center for Microscopy and Imaging Research, University of California, San Diego; **48.14** Alan Peters; **p. 1084** B.A.E./Alamy Stock Photo.

Chapter 49 **49.10** Tamily Weissman; **49.11** Larry Mulvehill/Corbis; **49.15** From: A functional MRI study of happy and sad affective states induced by classical music. M. T. Mitterschiffthaler et al. *Hum Brain Mapp*. 2007 Nov. 28(11):1150-62. Fig. 1; **49.18** Marcus E. Raichle, Washington University Medical Center. From research based on "Positron emission tomographic studies of the cortical anatomy of single-word processing". S.E. Petersen et al. *Nature* 331:585-589 (1988); **49.19** National Library of Medicine (NLM); **p. 1099** **Jarvis** Walter Oleksy/Alamy Stock Photo; **p. 1103** **Heberlein** Matthew Staley; **49.25** Martin M. Rotker/Science Source; **p. 1106** **microphone** Eric Delmar/Getty Images.

Chapter 50 **50.1** Kenneth Catania; **50.6** CSIRO Publishing; **50.7a** Michael Nolan/Robert Harding World Imagery; **50.7b** Grischa Georgiew/Panther Media/AGE Fotostock; **50.9** From: Richard Elzinga, *Fundamentals of Entomology*, 3rd ed. ©1987, p. 185. Reprinted by permission of Prentice-Hall, Upper Saddle River, NJ; **50.10** SPL/Science Source; **50.16a** APHIS Animal and Plant Health Inspection Service/USDA; **50.17** Steve Gschmeissner/Science Source; **50.21** Neitz Laboratories; **50.26** Clara Franzini-Armstrong; **50.27** H. E. Huxley; **50.34** Joe Quinn/Alamy Stock Photo; **50.39** Dave Watts/NHPA/Science Source; **p. 1135** Terence Dawson; **Scientific Skills Exercise** Vance A. Tucker; **p. 1138** **hound** Dogs/Fotolia.

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